BACKEND · CROSS-BORDER COMMERCE · CHONGQING, CHINA

Xiaozhe Bai

Backend Software Engineer focused on cross-border e-commerce systems.

jeff2025white@gmail.com Upwork Intro video

01 ABOUT

Backend engineer with 7 years at Alibaba, building distributed systems for AliExpress's global cross-border business. The work centers on order and transaction systems, merchant open APIs, high-concurrency optimization, and multi-region deployment — keeping high-traffic services stable through performance tuning and fault isolation.

02 SKILLS

Java · Go · Microservices · MySQL · Redis · Kafka · RocketMQ · API gateway · Distributed locks · DB sharding · Python · FastAPI · PostgreSQL · LLM integration · OCR / computer vision · PHP · Laravel

03 EXPERIENCE

Backend engineer

Alibaba · AliExpress · 7 years

Owned backend services for order, settlement, and merchant open APIs on a global cross-border marketplace, from high-concurrency tuning to multi-region deployment. See order & settlement and open API gateway below.

04 EDUCATION & LANGUAGES

Shanghai Jiao Tong University

Bachelor of Applied Science (BASc), Computer Science · 2015 – 2019

Languages

Chinese — native or bilingual

English — fluent (written)

EMPLOYER WORK

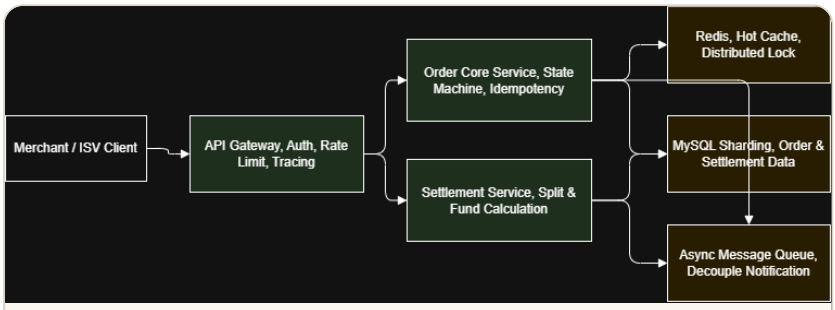
Selected work from a previous employer. Descriptions stay high-level and do not disclose confidential business data, internal metrics, or proprietary code.

Cross-border Order & Settlement Backend

Backend Software Engineer · AliExpress / Alibaba

Built a distributed order pipeline for a global cross-border marketplace. Implemented the order state machine, settlement splitting, idempotent control, and async notification for merchant services. Used MQ decoupling, DB sharding, distributed locks, and rate limiting to prevent overselling and duplicate orders. Tuned hot cache and slow SQL to stabilize peak traffic under flash sales, with multi-region global deployment.

Java Microservices MySQL Redis Apache Kafka



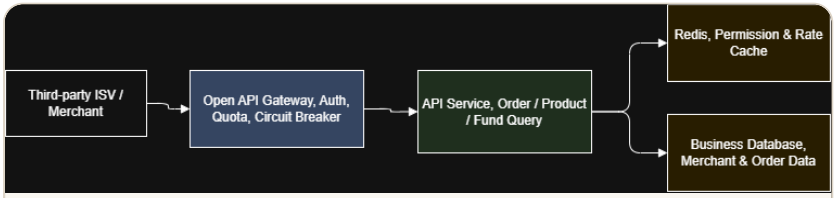
Architecture overview

Merchant Open API Gateway

Backend Software Engineer · AliExpress / Alibaba

Developed merchant-side open API services for third-party ISV integration. Designed authentication, permission control, throttling, tracing, and graceful degradation. Standardized API response specs and isolated service traffic to protect core business, so merchants could query order, product, and fund data in batches reliably.

Go API gateway Microservices Redis Performance optimization



Gateway architecture

PERSONAL PROJECTS

Personal projects. I own the copyright. They do not use or disclose any former employer's confidential business information. Repositories are private; access can be granted on request.

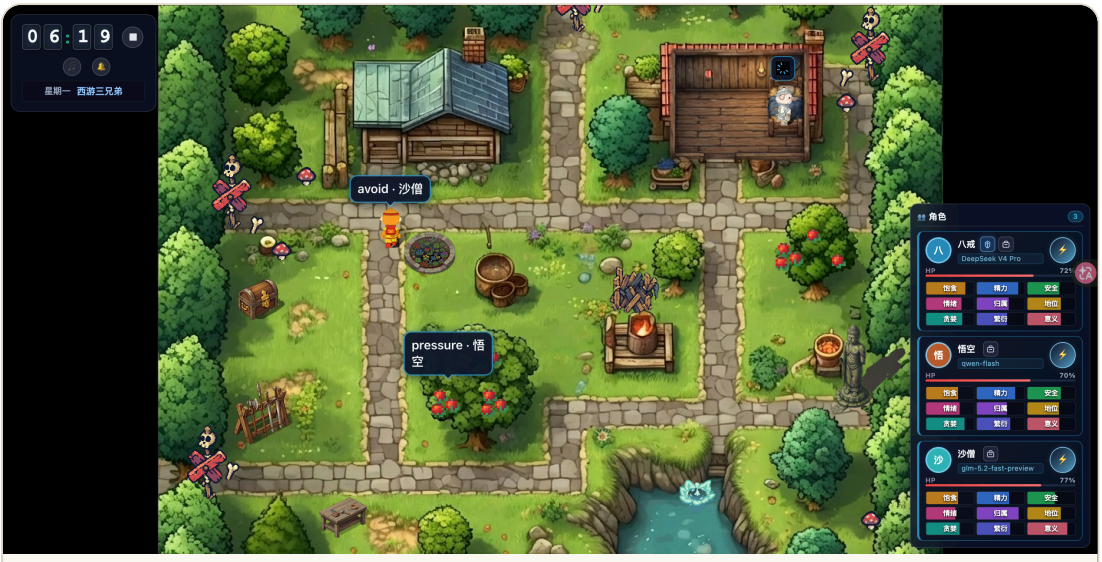
NPC2

Motive-driven generative agents · personal project

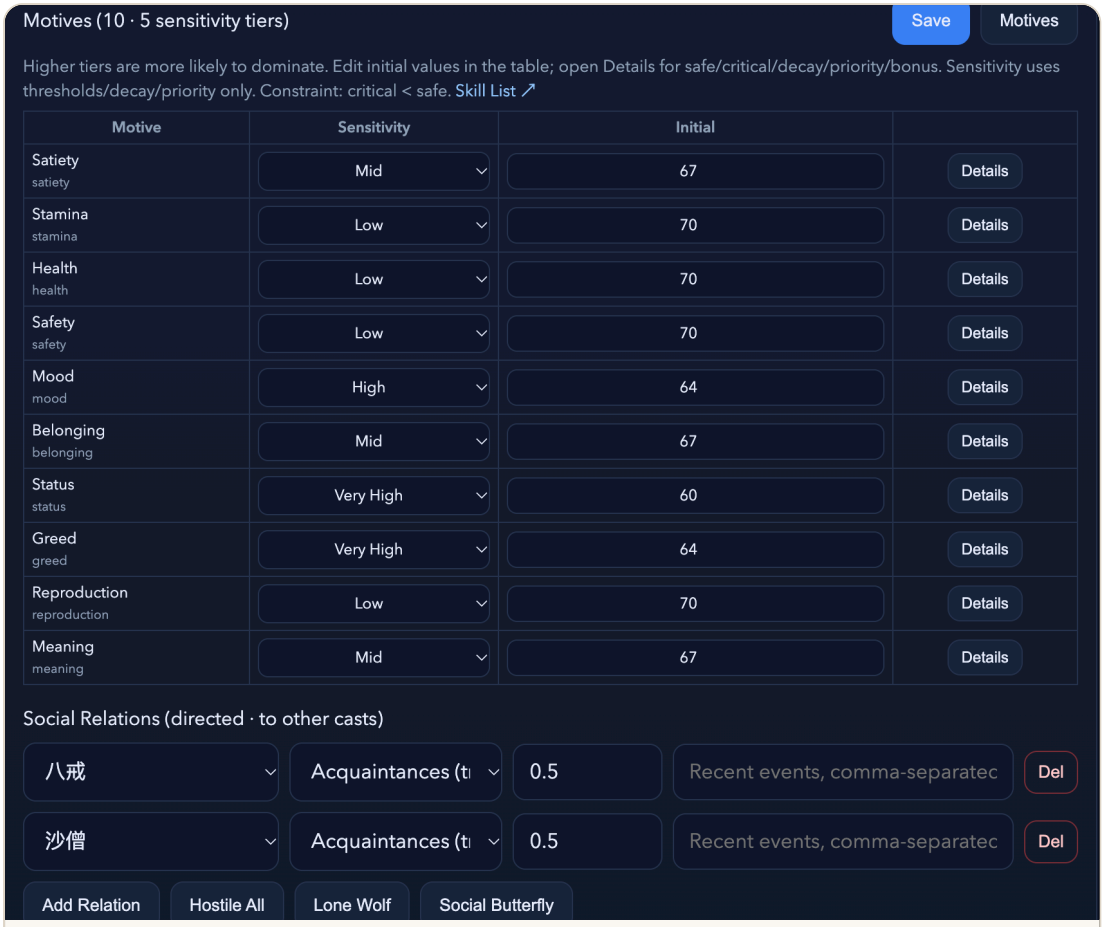
github.com/jeff2025white/npc2 Demo video Private

- Characters act from ten competing motives and a skill list, instead of a fixed script.
- Routine movement stays on a local pathfinder. Dialogue, conflict, and decisions call an LLM.
- Each character can use a different model. Motives, social ties, and skills are edited in a web console.

Python Django Phaser LLM routing



Live sandbox



Motive editor

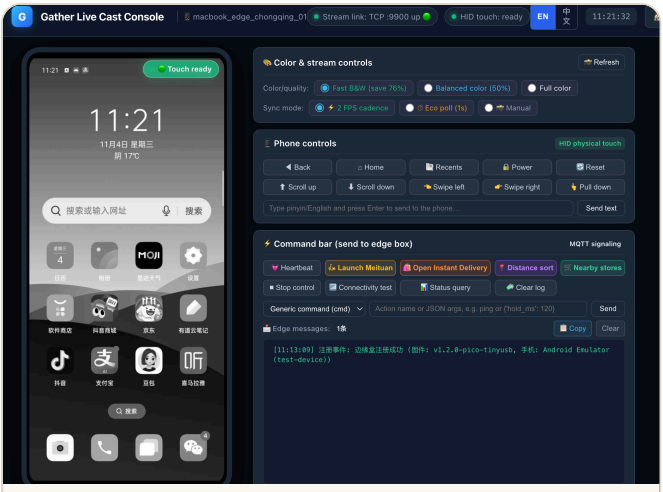
Gather + Flash

In-store capture and quick-commerce retail insight · personal project

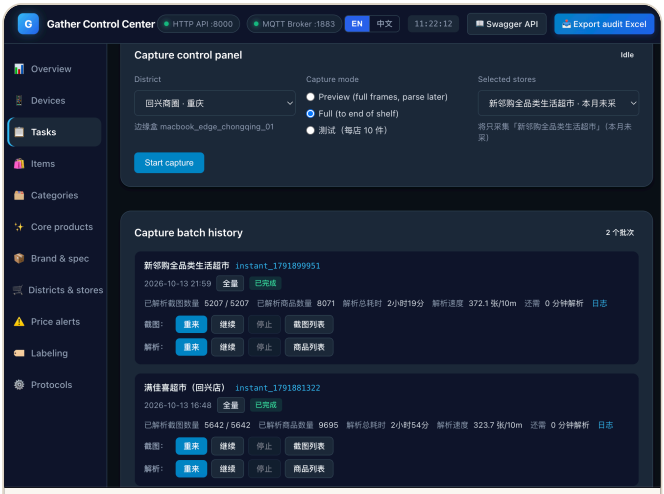
github.com/jeff2025white/gather github.com/jeff2025white/flash Private

- Gather drives a mirrored phone from an edge box: screen frames in, touch out through a custom RP2040 HID digitizer.
- A control center runs capture batches, parses shelf cards into price, sales, brand, and spec, and resumes large catalogs.
- Flash reads that catalog into a district board: assortment gaps, price position, and same-item sales against nearby stores.
- One capture batch parsed 5,207 screenshots in 2 h 19 min, about 370 frames every 10 minutes.

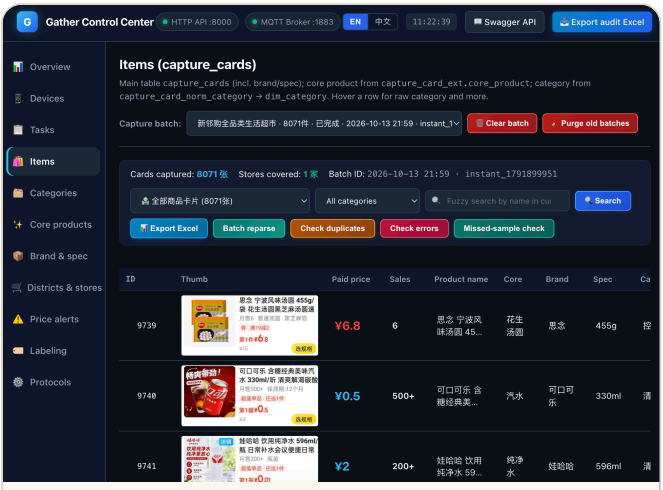
Python FastAPI PostgreSQL SQLite PaddleOCR MQTT RP2040 Kotlin



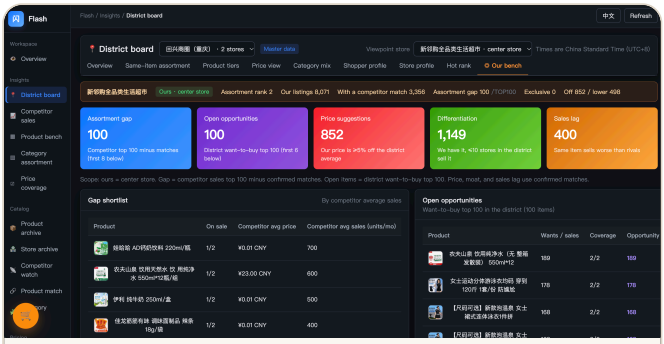
Gather · live cast



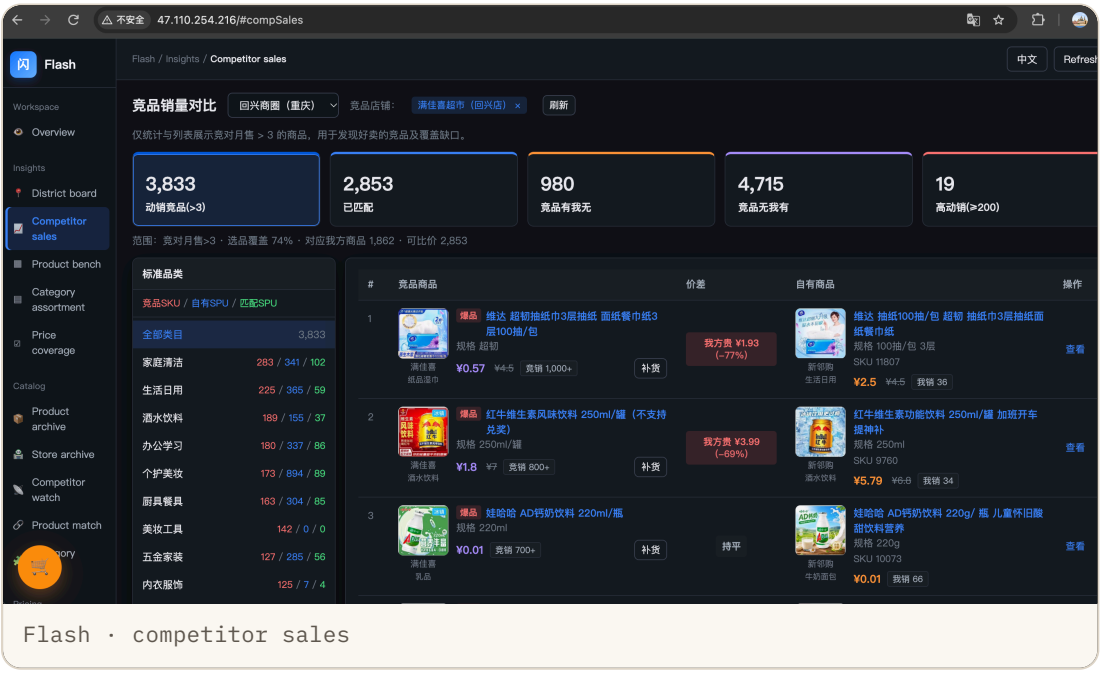
Gather · capture tasks



Gather · parsed cards



Flash · district board



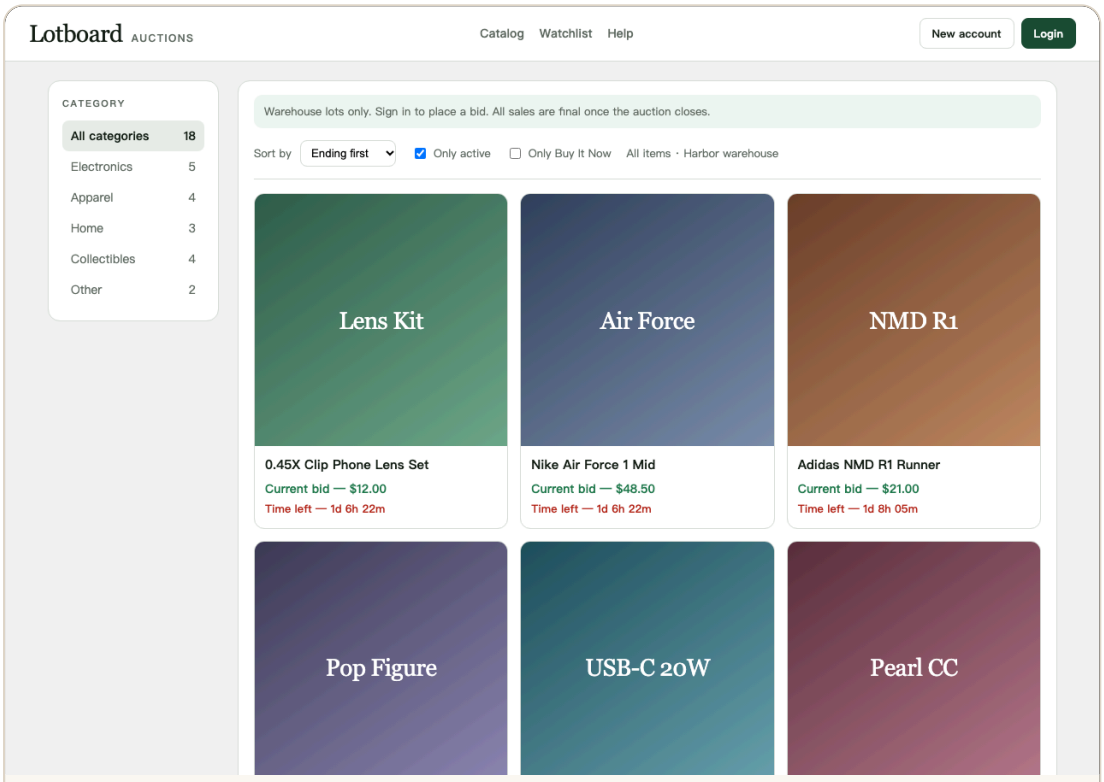
Flash · competitor sales

Online Auction Platform

Production-oriented timed bidding marketplace · personal project

- Built a full Laravel auction stack for warehouse-style lots: catalog, timed bidding, Buy It Now, bid history, and auth-gated bidding.
- Enforced safe concurrent bids with DB transactions and optimistic locking, plus increment rules (fixed step or percent of current bid).
- Shipped category filters, sort modes (ending soon, price, activity), multi-image lots, and a scheduler that closes auctions and selects winners.
- Hardened for production with policies/CSRF, rate-limited bid endpoints, queued notifications, Redis-backed cache/queues, and role-based admin for listings.

`Laravel` `PHP` `MySQL` `Redis` `Queues` `Scheduler` `Blade`



Auction catalog